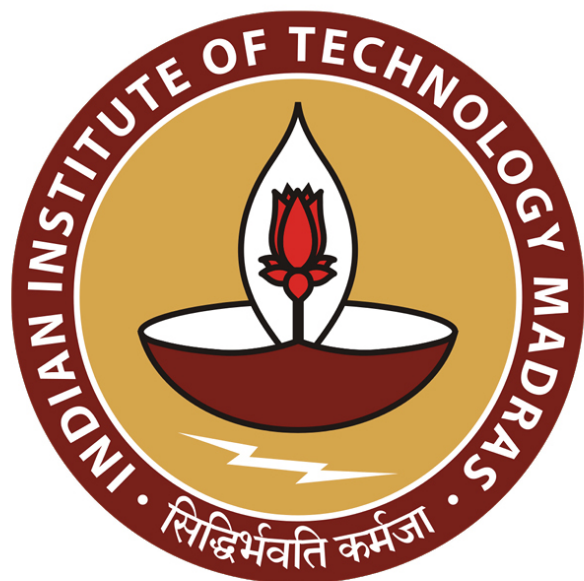
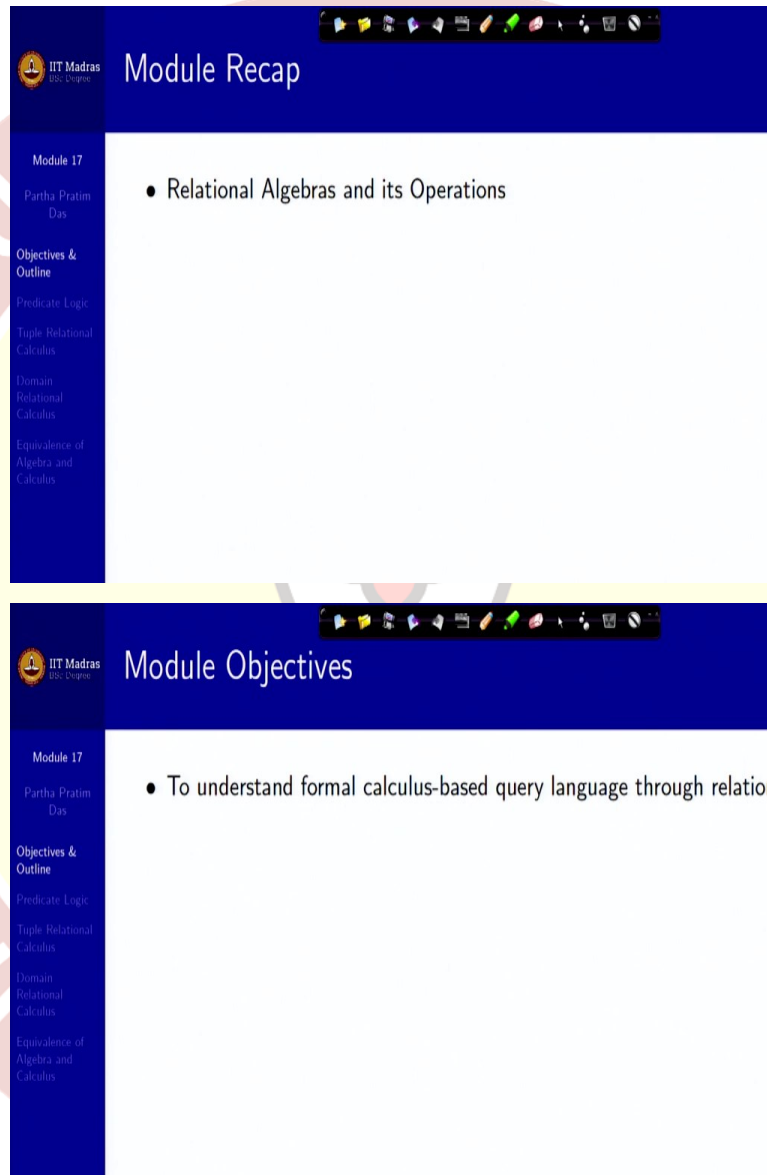




IIT Madras
ONLINE DEGREE

Database Management Systems
Professor Partha Pratim Das
Department of Computer Science and Engineering
Indian Institute of Technology, Kharagpur
Formal Relational Query Languages/2

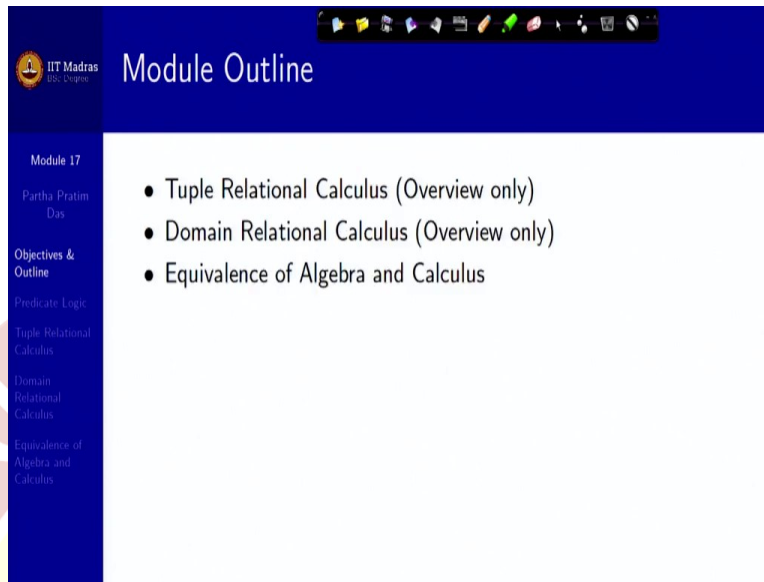
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The image displays two screenshots of the IIT Madras online BSc program interface. The top screenshot shows the 'Module Recap' slide for Module 17, listing topics: Relational Algebras and its Operations, Predicate Logic, Tuple Relational Calculus, Domain Relational Calculus, and Equivalence of Algebra and Calculus. The bottom screenshot shows the 'Module Objectives' slide for Module 17, listing the objective: To understand formal calculus-based query language through relation.

Welcome to Module 17 of Database Management Systems course in IIT Madras online BSc program. From the last module, we have started a little formal discussion about the foundational mathematics of relational database. We have discussed about relational algebra and its operation in some depth.

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Module Outline

Module 17
Partha Pratim Das

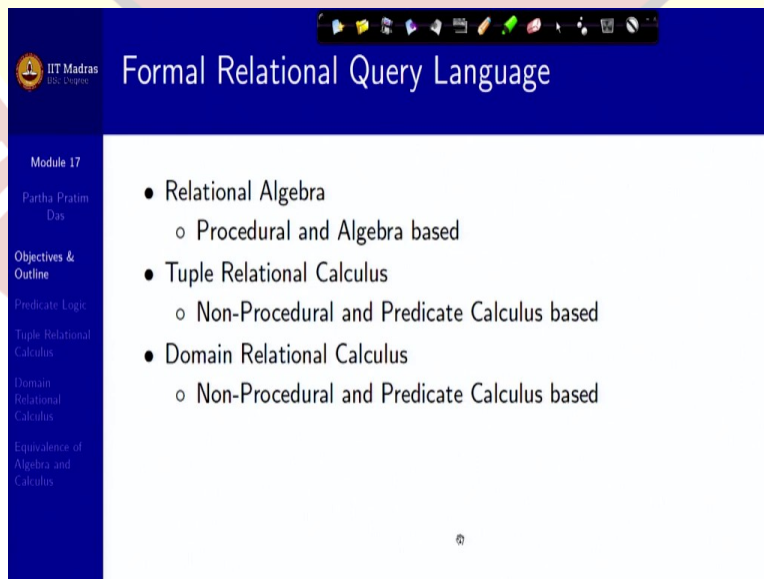
Objectives & Outline

- Predicate Logic
- Tuple Relational Calculus
- Domain Relational Calculus
- Equivalence of Algebra and Calculus

- Tuple Relational Calculus (Overview only)
- Domain Relational Calculus (Overview only)
- Equivalence of Algebra and Calculus

We will continue on that stream, but just take a quick look into the calculus-based query languages through the equivalence with relational algebra, we do not intend to really go deep on the calculus because first of all for two reasons one is the calculus needs more in-depth foundation on the results of predicate calculus, because they are based on predicate calculus and the other is they are all equivalent to relational algebra. So, just following one model on which particularly SQL is based would be quite okay for our future discussions.

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Formal Relational Query Language

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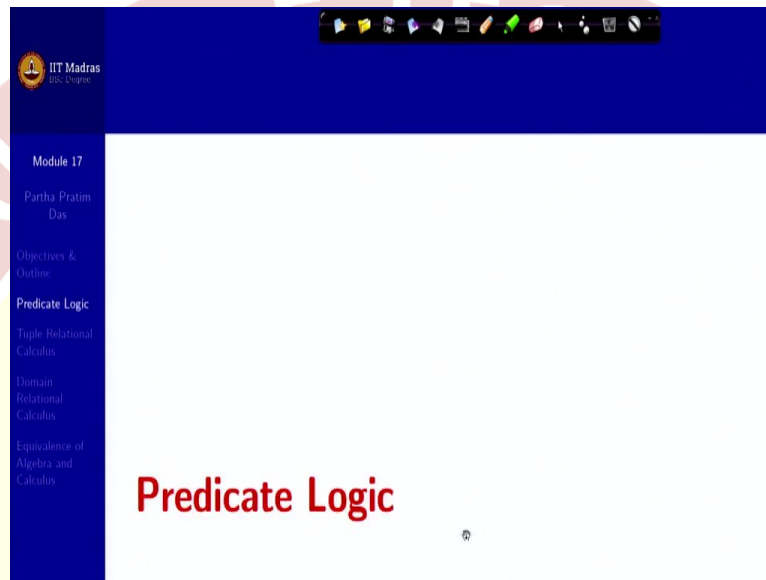
Objectives & Outline

- Predicate Logic
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- Domain Relational Calculus
- Equivalence of Algebra and Calculus

- Relational Algebra
 - Procedural and Algebra based
- Tuple Relational Calculus
 - Non-Procedural and Predicate Calculus based
- Domain Relational Calculus
 - Non-Procedural and Predicate Calculus based

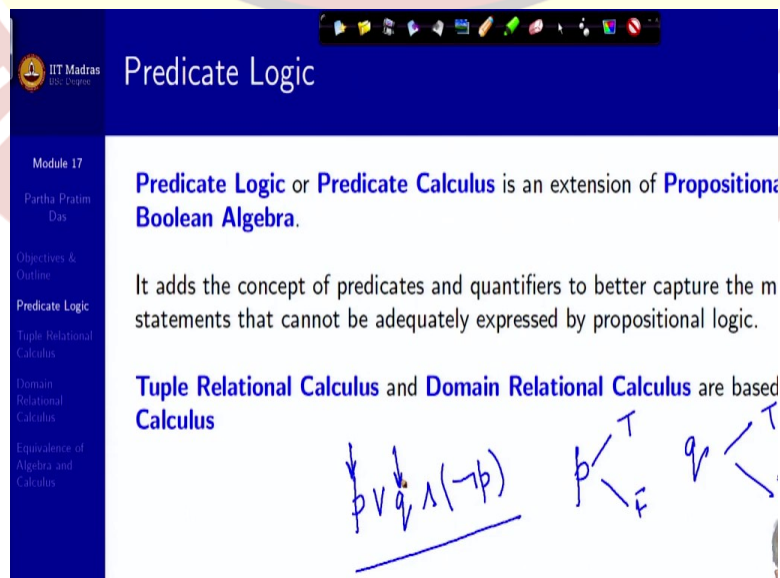
But it will be good for you all to know that these kind of models exist and what is their interrelationships with between themselves. So, the first one in the formal relational query language we have already done the second is what you get second and you know, just glimpse of the third will take.

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Now, before I get into tuple relational calculus, the first language or first model, let me just take a quick look at predicate logic because I am not sure how much of predicate logic you are familiar with, but few basic definitions are required for the entire of this calculus.

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So, predicate logic or predicate calculus, these are 2 names interchangeably used in some case places is called logic some place is called calculus, is an extension of propositional logic or propositional calculus or what more simply is called Boolean algebra. So, what we have in Boolean algebra?

In Boolean algebra, we have say variables, Boolean variables or propositions, which can be either true or false either true or false. And we have a few operators, three operators; say something like this and, or, not and we using them we can write any number of formula and once the truth value is assigned to these Boolean variables, we evaluate them to get a final truth result.

So, that is the basic formulation of Boolean algebra or propositional logic. So, you have propositions or variables which take true or false value and based on that you have a result. When it comes to predicate calculus, you have, you add a concept you still have propositions, but you add the concept of predicate and quantifiers to better capture the meaning of the statements which cannot be captured adequately appropriately precisely in terms of the propositional logic.

So, that is the extension. So, that is these are the key things to know, predicate and quantify plus the propositions A, B, C, D all variables which could be true or false. Now, tuple relational calculus and the domain calculus are completely based on predicate calculus. They are basically certain forms of predicate calculus.

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Predicate

- Consider the statement, " x is greater than 3". It has two parts. The variable x , is the subject of the statement. The second part, "is greater than 3", is the predicate. It refers to a property that the subject of the statement can have.
- The statement " x is greater than 3" can be denoted by $P(x)$ where P is the predicate "is greater than 3" and x is the variable.
- The predicate P can be considered as a function. It tells the truth value of the statement $P(x)$ at x . Once a value has been assigned to the variable x , $P(x)$ becomes a proposition and has a *truth* or *false* value.
- In general, a statement involving n variables $x_1, x_2, x_3, \dots, x_n$ can be denoted by $P(x_1, x_2, x_3, \dots, x_n)$. Here P is also referred to as an n -place predicate or n -ary predicate.

$P(2)$ $P(5)$

Let us take a quick look at what these predicates and quantifiers are, you may know but it is always good to discuss for completeness. So, suppose I make a statement that x is greater than 3. If you look into the statement, there are 2 things in this; one is the variable x , I do not know what its value is, what is the variable's normal value which can take some number 17, 23, 112 and so on. And the second part is a property, second part is actually a property. So, it says that is greater than 3. So, I have a variable and I have a property. Now, this part the property is called the predicate. It refers to the property that the subject of the statement x can have, may have or may not have.

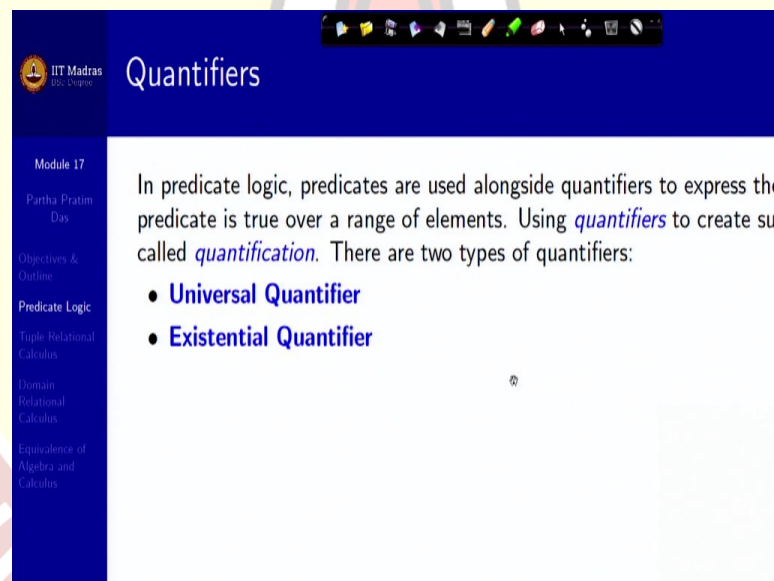
So, what I do is we denote this typically in this kind of a notation, where P is the predicate such that and x is the subject of that predicate x is a variable. So, that this predicate once I put a value to x this predicate is evaluated and becomes a proposition. So, $P(x)$ is a predicate but $P(5)$ is a proposition which has a value true or false. $P(2)$ is a proposition which has a value true or false.

Naturally if my predicate $P(x)$ is x greater than 3 then $P(5)$ the proposition $P(5)$ is true the proposition $P(2)$ is false. Alternatively, you can think about a predicate as a function because it tells you the truth value of $P(x)$ at the assumed value of x . So, you have predicates, you can again write conjunction, disjunction, negation with these predicates, you have a predicate calculus expression.

Once you put values truth value, once you put the different variable values x, y, z whatever. They become propositions. So, it becomes now from a predicate calculus formula it becomes a propositional calculus formula, a Boolean expression which will evaluate by the rules of Boolean expression and you finally have a truth value, the difference being that now, you are associating properties with variables which are not necessarily Boolean which are not necessarily binary, which can come from any domain and can have any property and a combination of those keeps you to capture the truth.

So, this obviously subsumes the entire of the propositional calculus but has lot more. Now, suddenly it is not limited to one variable. So, I can have any number of n number of variables for a predicate and I can denote them and if a variable if a predicate does not have a variable, it is a proposition.

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IIT Madras

Quantifiers

Module 17
Partha Pratim Das

Objectives & Outline
Predicate Logic
Tuple Relational Calculus
Domain Relational Calculus
Equivalence of Algebra and Calculus

In predicate logic, predicates are used alongside quantifiers to express the predicate is true over a range of elements. Using *quantifiers* to create such called *quantification*. There are two types of quantifiers:

- **Universal Quantifier**
- **Existential Quantifier**

The predicate logic also has alongside the predicates they have quantifiers. Quantifiers are used to talk about properties that hold over the entire domain of discourse in a certain way. Now, I said x is greater than 3. So, what is the domain of discourse for x , I can say it is a natural number. It is a natural number. If I said it is a natural number, then it is the whole of the natural numbers. Now, there are certain properties, which I want to know, whether it holds, how it holds over the entire domain of discourse.

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Universal Quantifier

Module 17
Partha Pratim Das

Objectives & Outline
Predicate Logic
Tuple Relational Calculus
Domain Relational Calculus
Equivalence of Algebra and Calculus

Universal Quantification: Mathematical statements sometimes assert to be true for all the values of a variable in a particular domain, called the **domain**.

- Such a statement is expressed using universal quantification.
- The universal quantification of $P(x)$ for a particular domain is the proposition that asserts that $P(x)$ is true for all values of x in this domain.
- The domain is very important here since it decides the possible values.
- Formally, The universal quantification of $P(x)$ is the statement " $P(x)$ in the domain".
- The notation $\forall P(x)$ denotes the universal quantification of $P(x)$. Here $\forall$ is the universal quantifier. $\forall P(x)$ is read as "for all x $P(x)$ ".
- Example: Let $P(x)$ be the statement " $x + 2 > x$ ". What is the truth value of $\forall P(x)$?

Universal Quantifier

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Universal Quantifier

Module 17

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- Example: Let $P(x)$ be the statement " $x + 2 > x$ ". What is the truth value of $\forall x P(x)$?

So, the first quantifier is known as a universal quantifier where it has a property which has to hold over the, why is it universal, because it talks about a property that has to universally hold true over the whole universe, which means it must be true for all elements in the domain. Only then, that quantification is true. So, the way I say so, it is written in terms of this notation for all.

So, I said this is a quantified formula for all x $P(x)$. Now, $P(x)$ depending on the value of x $P(x)$ may be true or may be false individually. So, what is for all x $P(x)$, it means, if I take any value in the domain of discourse, I can specify that domain or it can be implicitly known if I specify any value in that domain. And for that, if $P(x)$ is true, then this entire statement is true. This becomes a new predicate.

So, I say that, well, I am looking at a statement $x+2 > x$. If that is my $P(x)$ does this hold? Obviously, no matter what you take, the value of x as any real number, natural number $x+2$ will be always greater than x . So, I say that for all x $P(x)$, I say my predicate $M(x)$ is x is mortal. Whether domain of discourse is a population of the world, human population of the world, what is for all x $M(x)$, it is true; which in proverbial terms we say men are mortal, means human beings are mortal. So that is a universal quantifier. Let us say that we were saying that earlier we saw that $P(x)$ is $x > 3$. Now, what is for all x $P(x)$?

Over the domain of discourse, natural numbers. This quantified formula is not true. Well, it is true for 4, true for 5, true for 1000. But is not true for 3, true for 2, true for 1 and so on.

So, for all x $P(x)$, $x > 3$ is not true. But if I say my domain of discourse is all positive numbers having two or more digits, let us say my domain of discourse is all positive numbers having two or more digits then $P(x)$, $x > 3$ for all x $P(x)$ is true. So, you can see that the truth is always with respect to certain domains. And with respect to that, we are saying that if it holds universally, then it is universal true otherwise it is false.

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Existential Quantifier

Existential Quantification: Some mathematical statements assert that element with a certain property. Such statements are expressed by existential quantification. Existential quantification can be used to form a proposition and only if $P(x)$ is true for at least one value of x in the domain.

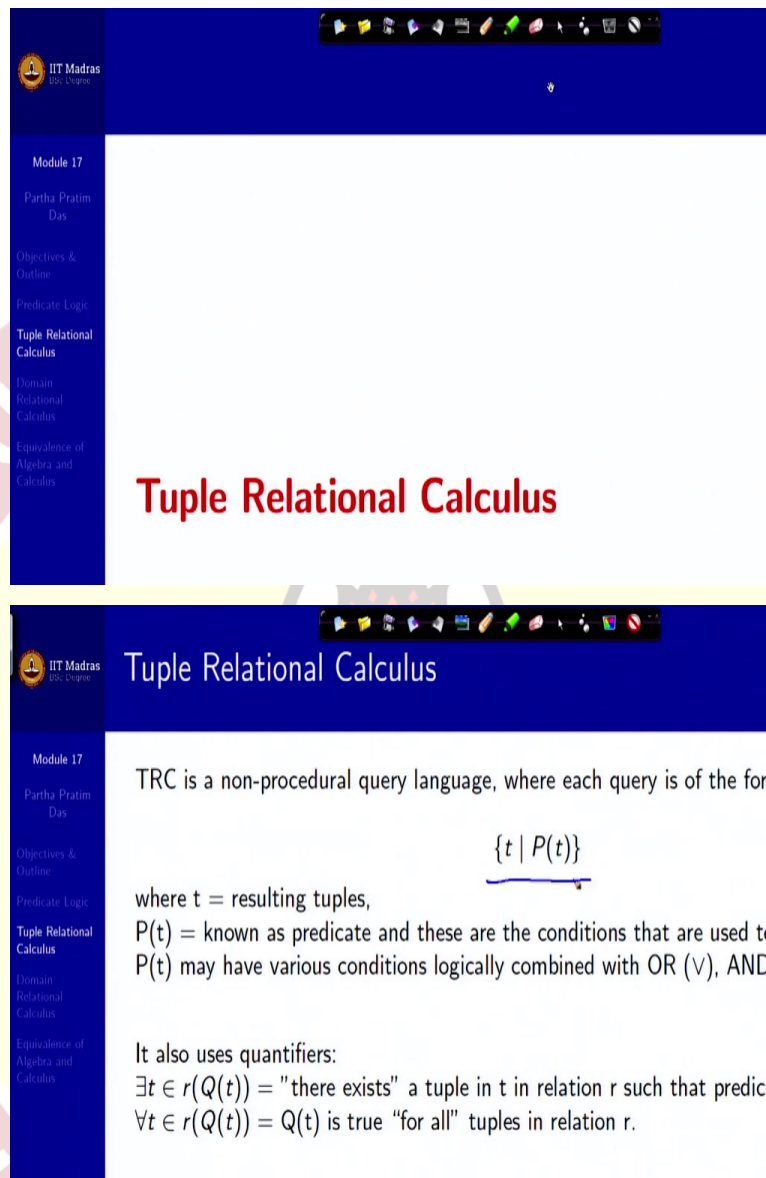
- Formally, the existential quantification of $P(x)$ is the statement "The element x in the domain such that $P(x)$ ".
- The notation $\exists P(x)$ denotes the existential quantification of $P(x)$. $\exists P(x)$ is read as "There is atleast one such x s
- Example: Let $P(x)$ be the statement " $x > 5$. What is the truth value $\exists x P(x)$?"

Solution: $P(x)$ is true for all real numbers greater than 5 and false for less than 5. So $\exists x P(x) = T$

So, let us move on, the other is existential quantifier which is on very similar lines where, where it says that does there exist at least one value in the domain of discourse where $P(x)$ is true, it is written in this way exist there exist x $P(x)$. So, if I if we go back to $P(x)$ being $x > 3$, then for all x $P(x)$ assuming the domain of discourse is natural numbers for all x $P(x)$ was false, but there exists x $P(x)$ is true because it is true for 4, true for 5 so at least one.

So, these are the two kinds of quantifications which we often need to model there are other kinds of quantifications also, but not so popular or rather to put it in different terms, they are expressible in terms of you know these quantification and the predicate. So, we move from propositional calculus to defining predicate and quantification and that model is what is the model for it.

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IIT Madras
Module 17
Partha Pratim Das
Objectives & Outline
Predicate Logic
Tuple Relational Calculus
Domain Relational Calculus
Equivalence of Algebra and Calculus

Tuple Relational Calculus

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Tuple Relational Calculus

TRC is a non-procedural query language, where each query is of the form

$$\{t \mid P(t)\}$$

where t = resulting tuples,
 $P(t)$ = known as predicate and these are the conditions that are used to
 $P(t)$ may have various conditions logically combined with OR ($\vee$), AND

It also uses quantifiers:
 $\exists t \in r(Q(t))$ = "there exists" a tuple in t in relation r such that predicate
 $\forall t \in r(Q(t)) = Q(t)$ is true "for all" tuples in relation r .

Is what the tuple relational calculus is. So, in tuple relational calculus all that we do is very simple we say that the formula or the relation is basically a set of t , tuples t where $P(t)$ where $P(t)$ holds good, where this predicate holds good and predicate may have various conditions combinations of and, or, not, it could have quantifiers and so on so forth.

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Predicate Calculus Formula

Module 17
Partha Pratim Das
Objectives & Outline
Predicate Logic
Tuple Relational Calculus
Domain Relational Calculus
Equivalence of Algebra and Calculus

- a) Set of attributes and constants
- b) Set of comparison operators: (e.g., $<$, $\leq$, $=$, $\neq$, $>$, $\geq$)
- c) Set of connectives: and ($\wedge$), or ($\vee$), not ($\neg$)
- d) Implication ($\Rightarrow$): $x \Rightarrow y$, if x is true, then y is true
 $x \Rightarrow y \equiv \neg x \vee y$
- e) Set of quantifiers:
 - $\exists t \in r(Q(t)) \equiv$ "there exists" a tuple in t in relation r such that predicate Q is true
 - $\forall t \in r(Q(t)) \equiv Q$ is true "for all" tuples t in relation r

So, so, that is the basic idea. So, these are the different component of the predicate calculus formula attributes would be there, constants would be there, comparison operators, connectives and, or, not implication x implies y which actually means not x or y and quantification. So, these are the possible formula for the tuple relational calculus.

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TRC Example

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Objectives & Outline
Predicate Logic
Tuple Relational Calculus
Domain Relational Calculus
Equivalence of Algebra and Calculus

Fname	Lname	Age	Course
David	Sharma	27	DBMS
Aaron	Lilly	17	JAVA
Sahil	Khan	19	Python
Sachin	Rao	20	DBMS
Varun	George	23	JAVA
Simi	Verma	22	JAVA

Q.1 Obtain the first name of students whose age is greater than 21.

Solution:

$\{t.Fname \mid Student(t) \wedge t.age > 21\}$

Fname
David
Varun
Simi

Couple of quick examples, say we are talking about obtain; there is a student's database here obtain the first name of the students whose age is greater than 21. So, what we are looking for is say if t is the relation then we are looking for the first name is $t.Fname$. What will have to happen

what is the domain of discourse? The truth depends on that; the domain of discourse is a relation student; student t student t is a predicate such that if t is a tuple of the student relation then student t is true otherwise it is false.

So, this will make sure that this is equivalent to saying in different terms as okay this is already written here let t is an element of truth you could be writing in this way and age is greater than So, $t.age$ is greater than 21. So, that defines this answer to this query, you could write it as a set member, you could also write it like this that in a set of tuples t set of names t such that you are there exist a tuple s with age greater than 21 and t name equal to S name in student. There is different ways of writing the formula means practically the same thing.

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TRC Example (2)

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Objectives & Outline
Predicate Logic
Tuple Relational Calculus
Domain Relational Calculus
Equivalence of Algebra and Calculus

Consider the relational schema
student(rollNo, name, year, courseId)
course(courseId, cname, teacher)

Q.2 Find out the names of all students who have taken the course name 'DBMS'.

- $\{t \mid \exists s \in \text{student} \exists c \in \text{course} (s.\text{courseId} = c.\text{courseId} \wedge c.\text{cname} = \text{'DBMS'} \wedge t.n$
- $\{s.\text{name} \mid s \in \text{student} \wedge \exists c \in \text{course} (s.\text{courseId} = c.\text{courseId} \wedge c.\text{cname} = \text{'DBMS'})\}$

Q.3 Find out the names of all students and their rollNo who have taken the course name 'DBMS'.

- $\{s.\text{name}, s.\text{rollNo} \mid s \in \text{student} \wedge \exists c \in \text{course} (s.\text{courseId} = c.\text{courseId} \wedge c.\text{cname} = \text{'DBMS'})\}$
- $\{t \mid \exists s \in \text{student} \exists c \in \text{course} (s.\text{courseId} = c.\text{courseId} \wedge c.\text{cname} = \text{'DBMS'} \wedge t.\text{name} = s.\text{name} \wedge t.\text{rollNo} = s.\text{rollNo})\}$

So, everything that we want to query on can be in this way written in multiple predicate calculus formula. Find the names of all students who have taken the course name DBMS. So, what is that we want? That for every student there is a course taken by the student called DBMS. So, let us look at the second one say $s.name$ for the names. Domain of discourse has to be a student.

Then we say that there exists a course c having the same Id as what the student has taken you know, the domain is getting refined, courses that the student has taken is a $s.courseId$. So, there exists a course c having the same Id and that course has the name DBMS. And when you evaluate the c , you do it in course table.

You could look at it differently; s is a student c is a course. So, the courseId is same and the name is DBMS and the students name matches that is a result t. So, that is basically the way to express in terms of predicate calculus or in this case, the tuple relational calculus. There are other queries.

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TRC Example (3)

Module 17
Partha Pratim Das
Objectives & Outline
Predicate Logic
Tuple Relational Calculus
Domain Relational Calculus
Equivalence of Algebra and Calculus

Consider the following relations:
Flights(flno, from, to, distance, departs, arrives)
Aircraft(aid, aname, cruisingrange)
Certified(eid, aid)
Employees(eid, ename, salary)

Q.4. Find the eids of pilots certified for Boeing aircraft.

RA

$$\Pi_{eid}(\sigma_{aname='Boeing'}(Aircraft \bowtie Certified))$$

TRC

- $\{C.eid \mid C \in Certified \wedge \exists A \in Aircraft(A.aid = C.aid \wedge A.aname = 'Boeing')\}$
- $\{T \mid \exists C \in Certified \exists A \in Aircraft(A.aid = C.aid \wedge A.aname = 'Boeing' \wedge T.eid = C.eid)\}$

TRC Example (3)

Module 17
Partha Pratim Das
Objectives & Outline
Predicate Logic
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Certified(eid, aid)
Employees(eid, ename, salary)

Q.4. Find the eids of pilots certified for Boeing aircraft.

RA

$$\Pi_{ename}(\Pi_{eid}(\sigma_{aname='Boeing'}(Aircraft \bowtie Certified)) \bowtie Employees)$$

TRC

- $\{C.eid \mid C \in Certified \wedge \exists A \in Aircraft(A.aid = C.aid \wedge A.aname = 'Boeing')\}$
- $\{T \mid \exists C \in Certified \exists A \in Aircraft(A.aid = C.aid \wedge A.aname = 'Boeing' \wedge T.eid = C.eid)\}$

Let us look at this one. So, there are flights, flight number, starting point, ending point, distance, departure time, arrival time, there is aircraft with ID, aircraft name, cruising range, how far it can go, the certified so there are employees which are certified on aircrafts and the employee details, these are the four tables.

So the queries is find the eids, employee ID of pilots, certified for Boeing aircraft, find the employee ID of pilot, this is your final result, certified for Boeing aircraft. So, you need the employee ID certified. So, you need a certified, you need the Boeing. So, all these will be involved.

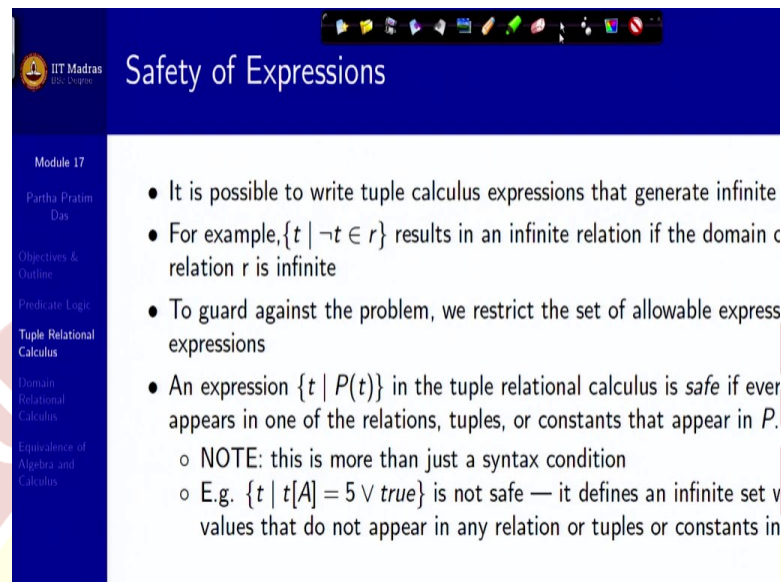
Now, the first thing you need is how do you say? How do you find what are the employee IDs that are available for the Boeing aircraft that are certified for the Boeing aircraft? Naturally, these two will have to come together. And what works for that, if you just think the relational algebra way, the natural join between aircraft and certified.

This will give you a relationship between employee ID and aircraft name, which could have Boeing. So now you filter this for Boeing. You are just interested in the Boeing part of it, nothing else. No other aircrafts. So, filter, select on Boeing. So, whatever will remain is the employee ID of those who are certified to fly eid. Here you have been asked to find the employee ID.

If you are asked to find the name, then what you will have to do? You'll have to again, because the name is available in the employees, you'll get the names. So, that is if you look at it in a relational way. Let us look at it in the calculus. It is very simple. In calculus actually, it is easier because what you are looking for are the certified employee IDs, c.eid. c has to be certified, has to come from there.

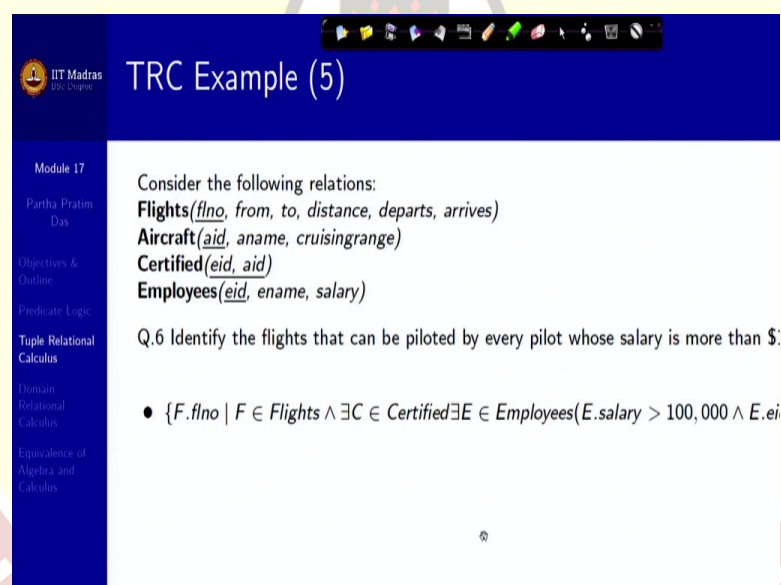
A has to be an aircraft which has a name Boeing and the certification of the aircraft ID available in the aircraft table and the aircraft ID available in the certification table has to match. That gives you the entire query. So, this is how, I mean, you need to, you need to practice a lot to be able to write this.

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Safety of Expressions

- It is possible to write tuple calculus expressions that generate infinite
- For example, $\{t \mid \neg t \in r\}$ results in an infinite relation if the domain relation r is infinite
- To guard against the problem, we restrict the set of allowable expressions
- An expression $\{t \mid P(t)\}$ in the tuple relational calculus is *safe* if ever appears in one of the relations, tuples, or constants that appear in P .
 - NOTE: this is more than just a syntax condition
 - E.g. $\{t \mid t[A] = 5 \vee \text{true}\}$ is not safe — it defines an infinite set v values that do not appear in any relation or tuples or constants in



TRC Example (5)

Consider the following relations:

- Flights**(flno, from, to, distance, departs, arrives)
- Aircraft**(aid, aname, cruisingrange)
- Certified**(eid, aid)
- Employees**(eid, ename, salary)

Q.6 Identify the flights that can be piloted by every pilot whose salary is more than \$:

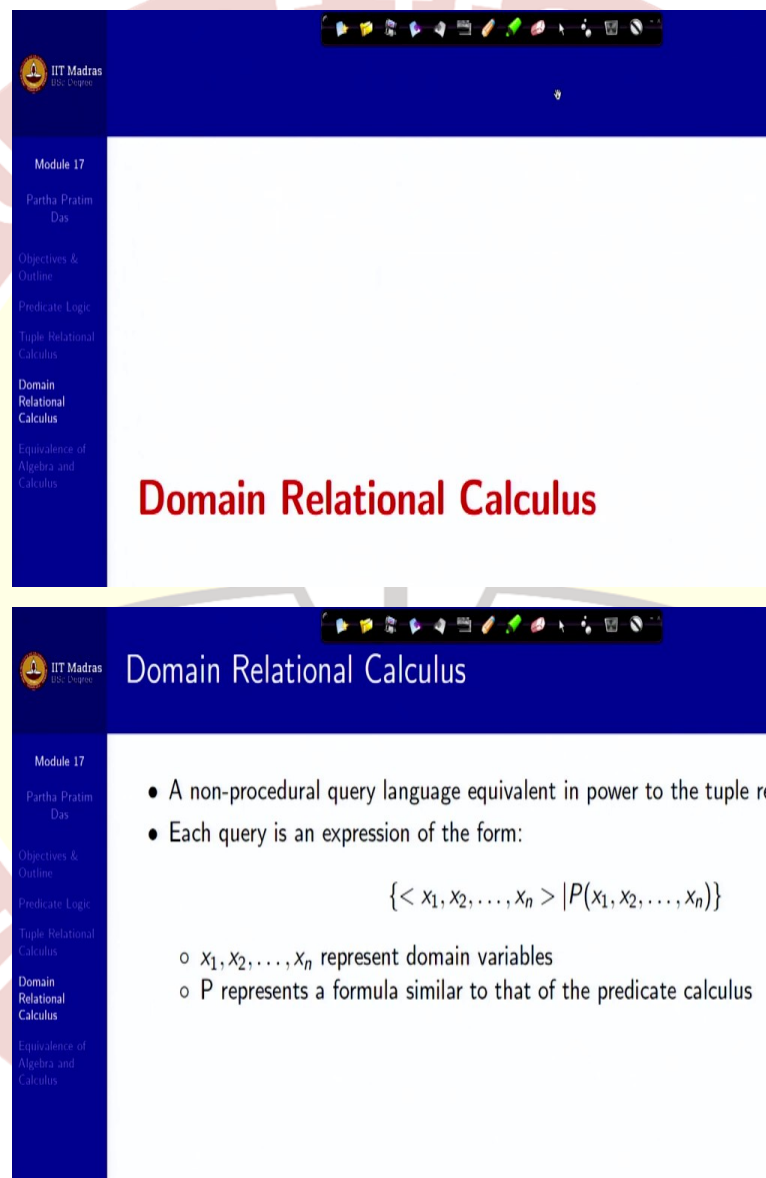
- $\{F.flno \mid F \in \text{Flights} \wedge \exists C \in \text{Certified} \exists E \in \text{Employees}(E.salary > 100,000 \wedge E.ei$

There are couple of things examples given round here, we can take them up in tutorials also. Now, so that is tuple relational calculus, it is just logical thinking nothing other than that. One point I would like to mention is before I close is that the you have to make sure that expressions are safe.

What does it mean, expression is safe, it means that the result of your expression cannot be infinite, it has to be finite, but so, you would say how can that happen? How come I am working with finite sets? How can it suddenly become infinite? It can because for example, I say that this is my predicate, set of tuples t such that t does not belong to relation r .

Now, that anything in the world can belong to, not belong to relation r . So, you have to be careful about that at least, if you have any attribute in the relation which has an infinite domain, then you must be careful to avoid that. Normally, it will not happen when you are working within SQL kind of situation because it is well taken care of within the SQL itself.

(Refer Slide Time: 26:17)



The image shows two screenshots of a presentation slide from IIT Madras. The top screenshot is the title slide, and the bottom screenshot is the content slide.

Title Slide:

- Module 17
- Partha Pratim Das
- Objectives & Outline
- Predicate Logic
- Tuple Relational Calculus
- Domain Relational Calculus**
- Equivalence of Algebra and Calculus

Domain Relational Calculus

Content Slide:

- A non-procedural query language equivalent in power to the tuple rel.
- Each query is an expression of the form:
$$\{ \langle x_1, x_2, \dots, x_n \rangle \mid P(x_1, x_2, \dots, x_n) \}$$
 - $x_1, x_2, \dots, x_n$ represent domain variables
 - P represents a formula similar to that of the predicate calculus

The domain relational calculus, I will not go in in detail, as practically no difference between the tuple relational calculus and domain relational calculus except the notation. In the tuple relational calculus, you have variable for the entire tuple club together. In domain relational calculus, every attribute is a different variable. So, you have a Cartesian product of that, which is the, so if I have

an array tuple, then in domain relational, in, in tuple relational calculus, I call it t in domain relational calculus, I will call it the attributes x1, x2, xn and so, what becomes anywhere you use? So, it is just a different way of, you know, syntactically writing things, which makes it different conceptually, there is absolutely no difference between the two of them.

(Refer Slide Time: 27:09)

Equivalence of Algebra and Calculus

Equivalence of RA, TRC and DRC

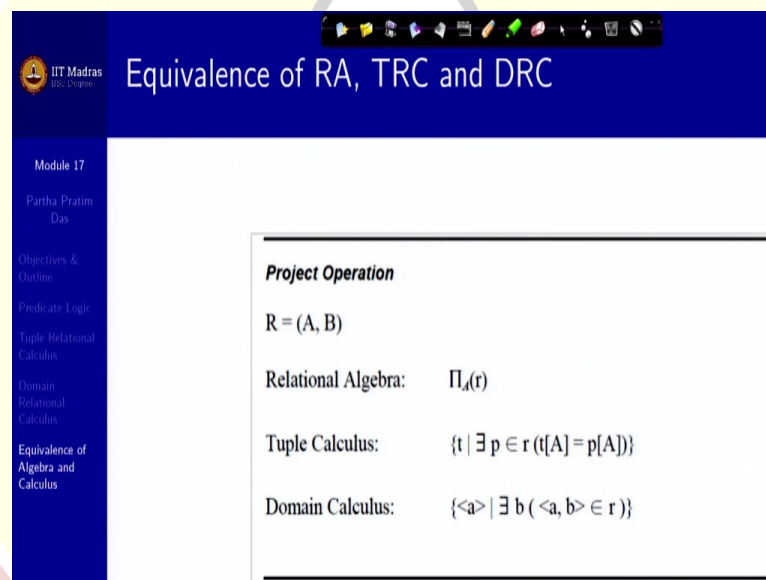
Select Operation	
$R = (A, B)$	
Relational Algebra:	$\sigma_{B=17}(r)$
Tuple Calculus:	$\{t \mid t \in r \wedge B = 17\}$
Domain Calculus:	$\{ \langle a, b \rangle \mid \langle a, b \rangle \in r \wedge b = 17 \}$

Now, finally the question is, I mean, the three models are the same. Now, in some cases, we have worked out, you can there is a formal proof for everything what the way you go is, is basically for every relational algebra operator. If you can show, how do you write the tuple calculus expression, then you will know that that is equivalent.

That is, it is similarly for every tuple calculus predicate that is conjunction, disjunction, negation quantification, if you can show what is equivalent in the relational algebra, then you will know that both of them are equivalent. So, just showing you an example here that this is a selection operator $\sigma_{B=17}$ on (r) .

So, we know that in tuple calculus, it can be written like this, and if the relation has two attributes in domain calculus, it can be written like this. Now, this is by example, you can try to prove it in general, I am not going into those because, we want to keep closer to practice. These are things that people have worked on and have proven long ago, and you can take that, take that fact.

(Refer Slide Time: 28:39)



Equivalence of RA, TRC and DRC

Module 17
Partha Pratim Das

Objectives & Outline
Predicate Logic
Tuple Relational Calculus
Domain Relational Calculus
Equivalence of Algebra and Calculus

Project Operation

$R = (A, B)$

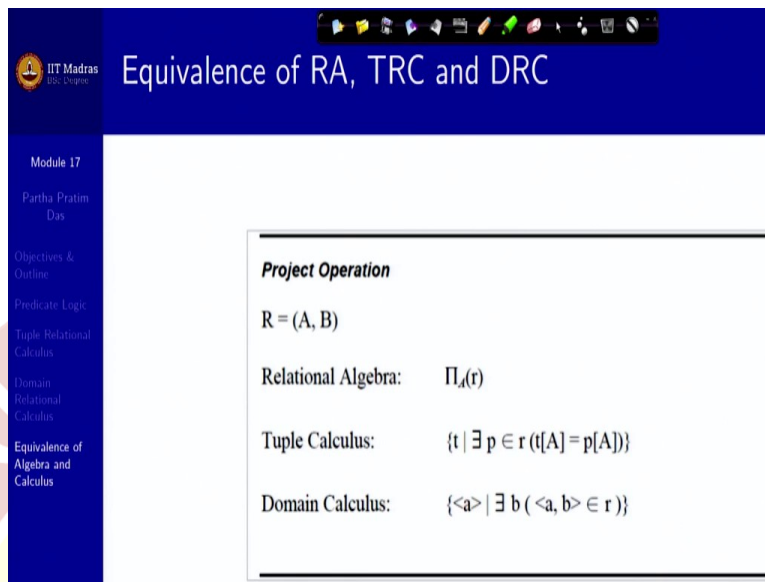
Relational Algebra: $\Pi_A(r)$

Tuple Calculus: $\{t \mid \exists p \in r (t[A] = p[A])\}$

Domain Calculus: $\{ \langle a \rangle \mid \exists b (\langle a, b \rangle \in r) \}$

So, all that I have done is for every operation, I have just given one example, to elucidate that, what is the kind of structure that it equivalently takes. So, this is a projection. So, when you take a projection, how you can write it in terms of tuple calculus and so on. But we would not go into the detail mathematical proof of this. The proof is given in the book if you want you can read it up.

(Refer Slide Time: 29:09)



The slide is titled "Equivalence of RA, TRC and DRC". It features a sidebar on the left with the IIT Madras logo and a list of topics: Module 17, Partha Pratim Das, Objectives & Outline, Predicate Logic, Tuple Relational Calculus, Domain Relational Calculus, and Equivalence of Algebra and Calculus. The main content area is titled "Project Operation" and shows the following:

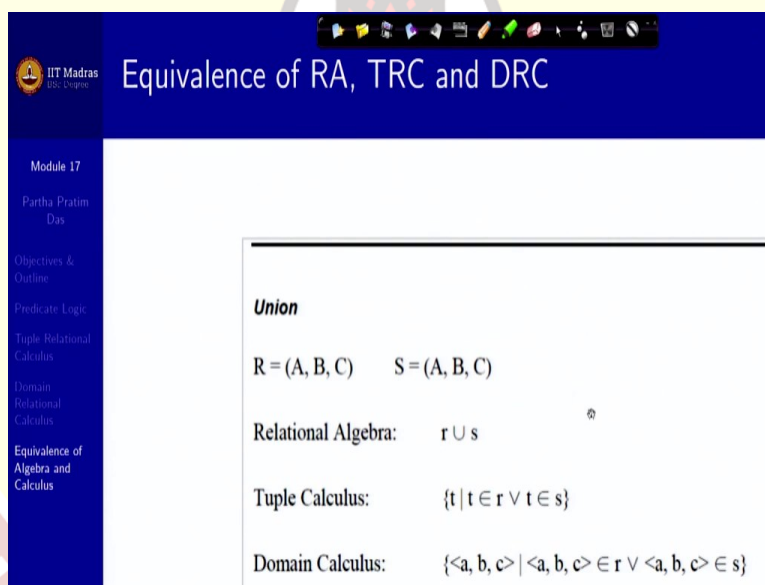
Project Operation

$R = (A, B)$

Relational Algebra: $\Pi_A(r)$

Tuple Calculus: $\{t \mid \exists p \in r (t[A] = p[A])\}$

Domain Calculus: $\{ \langle a \rangle \mid \exists b (\langle a, b \rangle \in r) \}$



The slide is titled "Equivalence of RA, TRC and DRC". It features a sidebar on the left with the IIT Madras logo and a list of topics: Module 17, Partha Pratim Das, Objectives & Outline, Predicate Logic, Tuple Relational Calculus, Domain Relational Calculus, and Equivalence of Algebra and Calculus. The main content area is titled "Union" and shows the following:

Union

$R = (A, B, C) \quad S = (A, B, C)$

Relational Algebra: $r \cup s$

Tuple Calculus: $\{t \mid t \in r \vee t \in s\}$

Domain Calculus: $\{ \langle a, b, c \rangle \mid \langle a, b, c \rangle \in r \vee \langle a, b, c \rangle \in s \}$



Equivalence of RA, TRC and DRC

Module 17

Partha Pratim Das

Objectives & Outline

Predicate Logic

Tuple Relational Calculus

Domain Relational Calculus

Equivalence of Algebra and Calculus

Cartesian/Cross Product

$$R = (A, B) \quad S = (C, D)$$

Relational Algebra: $r \times s$

Tuple Calculus: $\{t \mid \exists p \in r \exists q \in s (t[A] = p[A] \wedge t[B] = p[B] \wedge t[C] = q[C] \wedge t[D] = q[D])\}$

Domain Calculus: $\{ \langle a, b, c, d \rangle \mid \langle a, b \rangle \in r \wedge \langle c, d \rangle \in s \}$



Equivalence of RA, TRC and DRC

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Equivalence of Algebra and Calculus

Set Difference

$$R = (A, B, C) \quad S = (A, B, C)$$

Relational Algebra: $r - s$

Tuple Calculus: $\{t \mid t \in r \wedge t \notin s\}$

Domain Calculus: $\{ \langle a, b, c \rangle \mid \langle a, b, c \rangle \in r \wedge \langle a, b, c \rangle \notin s \}$

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Equivalence of RA, TRC and DRC

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Equivalence of Algebra and Calculus

Natural Join

$R = (A, B, C, D) \quad S = (B, D, E)$

Relational Algebra: $r \bowtie s$

$$\Pi_{r.A, r.B, r.C, r.D, s.E}(\sigma_{r.B=s.B \wedge r.D=s.D}(r \times s))$$

Tuple Calculus:

$$\{t \mid \exists p \in r \exists q \in s (t[A] = p[A] \wedge t[B] = p[B] \wedge t[C] = p[C] \wedge t[D] = p[D] \wedge t[E] = q[E] \wedge p[B] = q[B] \wedge p[D] = q[D])\}$$

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Equivalence of RA, TRC and DRC

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Division

$R = (A, B) \quad S = (B)$

Relational Algebra: $r \div s$

Tuple Calculus:

$$\{t \mid \exists p \in r \forall q \in s (p[B] = q[B] \Rightarrow t[A] = p[A])\}$$

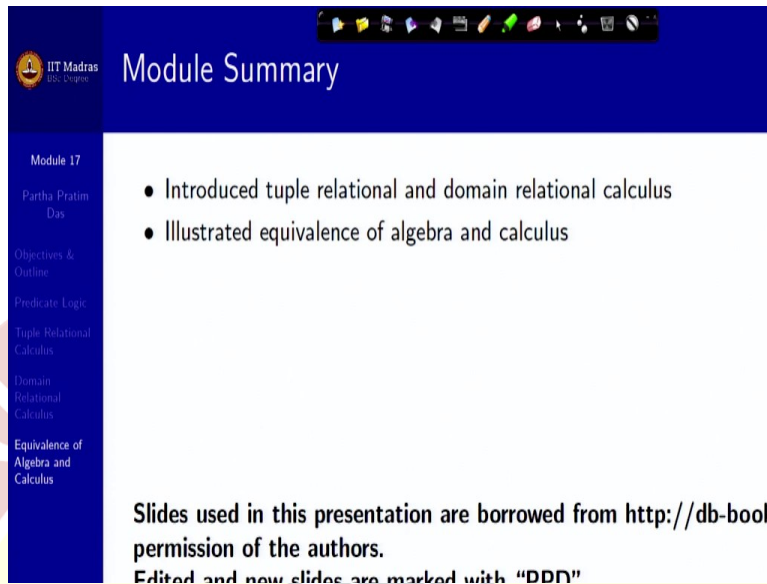
Domain Calculus:

$$\{ \langle a \rangle \mid \langle a \rangle \in r \wedge \forall \langle b \rangle (\langle b \rangle \in s \Rightarrow \langle a, b \rangle \in r) \}$$

But just go with the fact that there is equivalence for each and every operation and their combination between relational algebra and the predicates in the calculus. So, this is for union, set difference, you can even just see through that when you are reading through this slides, the Cartesian product everything natural join.

So, it looks a little cumbersome, when you write down natural join, but you know, you can always it is always possible to write down a natural join in tuple calculus, because it is possible to write a Cartesian product and the rest of it is just selection under different conditions. Division too, though division is not a fundamental operator. So, this is not necessary for the equivalence proof.

(Refer Slide Time: 30:00)



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Module Summary

Module 17
Partha Pratim Das

- Introduced tuple relational and domain relational calculus
- Illustrated equivalence of algebra and calculus

Objectives & Outline
Predicate Logic
Tuple Relational Calculus
Domain Relational Calculus
Equivalence of Algebra and Calculus

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Edited and new slides are marked with "DDDD"

So, with this, we have discussed the tuple relational domain relational calculus, the basic framework of that I said that I will be at the overview level, when primarily we will continue to use the relational algebra and we just mentioned an outline of how you can probably go around looking at the equivalence of them. So, we will close the module here now. Thank you very much for your attention and see you in the next module.